NEPAL COLLEGE OF INFORMATION TECHNOLOGY

Balkumari, Lalitpur

Affiliated to Pokhara University

ASSIGNMENT FOR COMPUTER GRAPHICS

notes.arpanadhikari7.com.np

ASSIGNMENT 3

Submitted by:
Name: Arpan Adhikari
Roll No.: 231309
BE-CE (3rd SEM)

Submitted to:
Saroj Shakya

Available on notes.arpanadhikari7.com.np

Arpan Adhikari
Roll No. 231309
Computer Graphics
Assignment 3

1) A point P(20,20) in a circular window with its center at (100,100) and a radius of 90 pixels is required to be mapped to a point in circular viewport having its at (50,50) and radius of 40 pixels. Where will the point P be placed after the transformation?

Sol'n

Radius of window (W_r) = 90 $\quad (x_{wc}, y_{wc}) = (100,100)$
Radius of viewport (V_r) = 40 $\quad (x_{vc}, y_{vc}) = (50,50)$

So, $S_x = \frac{V_r}{W_r} = \frac{40}{90} = 0.44$ $\quad S_y = \frac{V_r}{W_r} = 0.44$

Composite Matrix (C_m) = $T_{(x_{vc}, y_{vc})} \cdot S_{wv} \cdot T_{(-x_{wc}, -y_{wc})}$

notes.arpanadhikari7.com.np

$$= \begin{bmatrix} 1 & 0 & 50 \\ 0 & 1 & 50 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0.44 & 0 & 0 \\ 0 & 0.44 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -100 \\ 0 & 1 & -100 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.44 & 0 & 6 \\ 0 & 0.44 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

So, $P' = C_m \cdot P$

$$= \begin{bmatrix} 0.44 & 0 & 6 \\ 0 & 0.44 & 6 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 20 \\ 1 \end{bmatrix} = \begin{bmatrix} 14.8 \\ 14.8 \\ 1 \end{bmatrix}$$

Available on notes.arpanadhikari7.com.np

Hence, the point P(20,20) after being mapped to viewport will be P'(14.8, 14.8) ≈ P'(15, 15). Ans

Q2) A triangle with vertices A(5,2), B(4,1), C(6,1) is required to be rotated in a clockwise direction by 45 degrees about any arbitary point (4,4). Find out the final coordinate positions of the triangle after performing the desired transformation.

Sol

Here, $\theta = -45°$ (cw)

Now,

composite matrix $(C_m) = T_{(4,4)} \cdot R^{cw}_{\theta=-45°} \cdot T_{(-4,-4)}$

$$= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(-45) & -\sin(-45) & 0 \\ \sin(-45) & \cos(-45) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45 & \sin 45 & 0 \\ -\sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & -4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.707 & 0.707 & -1.66 \\ -0.707 & 0.707 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

Hence, $P' = C_m \cdot P$

$$= \begin{bmatrix} 0.707 & 0.707 & -1.66 \\ -0.707 & 0.707 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 4 & 6 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Available on notes.arpanadhikari7.com.np

$$= \begin{bmatrix} 3.29 & 1.87 & 3.29 \\ 1.87 & 1.87 & 0.46 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} 3 & 2 & 3 \\ 2 & 2 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Hence, the vertices of triangle after transformation are
A'(3,2), B'(2,2), C'(3,0).

Q3) Reflect a triangle A(1,0), B(3,1), C(1,2) about line $y = -2x + 5$ then scale it about a fined point P(10,10).

Soln

Here, $y = -2x + 5$

So, $m = -2$, $c = 5$

Or, $\theta = \tan^{-1}(-2)$

$\therefore \theta = -63.43^\circ$

Lets assume scaling factor as '2' in both directions.

$$C_m = S_{(2,2)} T_{(0,5)} \cdot R_{(cw)}^{-63.43^\circ} R_{fx} \cdot R_{(ccw)}^{63.43^\circ} \cdot T_{(0,-5)}$$

$$= \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 63.43^\circ & \sin 63.43^\circ & 0 \\ -\sin 63.43^\circ & \cos 63.43^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 63.43^\circ & -\sin 63.43^\circ & 0 \\ \sin 63.43^\circ & \cos 63.43^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{pmatrix}$$

Available on notes.arpanadhikari7.com.np

$$= \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -0.599 & -0.80 & 4 \\ -0.80 & 0.599 & -2.99 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1.19 & -1.60 & 8.00 \\ -1.60 & 1.19 & 4.00 \\ 0 & 0 & 1 \end{pmatrix}$$

So, $P' = C_m \cdot P$

$$= \begin{pmatrix} -1.19 & -1.60 & 8 \\ -1.60 & 1.19 & 4 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 6.81 & 2.83 & 3.61 \\ 2.4 & 0.39 & 4.78 \\ 1 & 1 & 1 \end{pmatrix}$$

Hence, Final coordinates are (7,2), (3,0), (4.4,5) Ans

Q4) A triangle with vertices A(5,2), B(4,1), C(6,1) is required to be reflected about an arbitary line $y = 2x+1$. Find out the final coordinate positions of the triangle after performing the desired transformation.

Soln

notes.arpanadhikari7.com.np

Given line, $y = 2x + 1$

So, $m = 2$

or, $\theta = \tan^{-1}(2)$

$\therefore \theta = 63.43°$

Composite Matrix $(C_m) = R_{\theta}^{63.43}{}_{ccw} \; R_{fx} \; R_{\theta}^{-63.43}{}_{cw}$

$$= \begin{bmatrix} \cos 63.43° & -\sin 63.43° & 0 \\ \sin 63.43° & \cos 63.43° & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Available on notes.arpanadhikari7.com.np

$$= \begin{bmatrix} -0.59 & 0.81 & 0 \\ 0.81 & 0.59 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, $P' = C_m \times P$

$$= \begin{bmatrix} -0.59 & 0.81 & 0 \\ 0.81 & 0.59 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 4 & 6 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -1.33 & -1.55 & -2.73 \\ 5.23 & 3.83 & 5.45 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\approx \begin{bmatrix} -1 & -2 & -3 \\ 5 & 4 & 5 \\ 1 & 1 & 1 \end{bmatrix}$$

Hence, vertices after transformation are A'(-1,5), B'(-2,4), C'(-3,5).

Q5) A triangle with vertices A(5,2), B(4,1), C(6,1) is required to be rotated by 45 degrees in CC direction about (i) origin and (ii) line $y = 5$.

Soln

i) Origin

Step 1 : Rotate the object by 45° in CCW.

$$R_{\theta = 45°, \text{ccw}} = \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The required component matrix is as,

Available on notes.arpanadhikari7.com.np

$$= \begin{bmatrix} 0.7071 & -0.7071 & 0 \\ 0.7071 & 0.7071 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Now, $\begin{bmatrix} 0.7071 & -0.7071 & 0 \\ 0.7071 & 0.7071 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 4 & 6 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$$= \begin{bmatrix} 2.1213 & 2.1213 & 3.5355 \\ 4.9497 & 3.5355 & 4.9497 \\ 1 & 1 & 1 \end{bmatrix}$$

(ii) $y = 5$ line :

Step1 : Translate the line $y = 5$ to x-axis.

Step2 : Rotate the object by 45° in ccw.

Step3 : Translate back the line along with rotated object to original position

The composite matrix is as,

$$C_m = T(0,5) \cdot R_{cw}^{45^\circ} \cdot T(0,-5)$$

$$C_m = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 45 & -\sin 45 & 0 \\ \sin 45 & \cos 45 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.7071 & -0.707 & 0 \\ 0.7071 & 0.7071 & 5 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0.7071 & -0.7071 & 3.53 \\ 0.7071 & 0.7071 & 1.46 \\ 0 & 0 & 1 \end{bmatrix}$$

Available on notes.arpanadhikari7.com.np

Now, The matrix coordinates after rotation is,

$$P' = \begin{bmatrix} 0.7071 & -0.7071 & 3.53 \\ 0.7071 & 0.7071 & 1.46 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & 4 & 6 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5.6513 & 5.6513 & 7.0655 \\ 6.4097 & 4.9955 & 6.4097 \\ 1 & 1 & 1 \end{bmatrix}$$

Hence, the points are A'(6.6), B'(6,5), C'(7,6).

Q6) Clip a line with end point coordinates A(-1,6), B(5,-8) against a clip window with its lower left corner at (-2,-5) and upper right corner at (4,8) using cohen-sutherland algorithm.

Soln

step 1 : Establish region codes for line end points.

A(-1,6)			B(5,-8)		
L : -1 < -2	F	'0'	L : 5 < -2	F	0
R : -1 > 4	F	'0'	R : 5 > 4	T	1
B : 6 < -5	F	'0'	B : -8 < -5	T	1
T : 6 > 8	F	'0'	T : -8 > 8	F	0

So, Bits = 0000 Bits for B = 0110.

So, this is not the case of full visibility.

Available on notes.arpanadhikari7.com.np

step 2: Perform AND operation of region codes.

0000
0110
0000

so, this is the case of partial visibility.

For B,

Here, slope(m) = $\frac{y_2 - y_1}{x_2 - x_1} = \frac{-8-6}{5-(-1)} = \frac{-14}{6}$

Step 3: Determine intersection points of line with window boundaries.

For B(5, -8),

For bit 2,

$y = y_{wmax} = 4$

and, $x = x_1 + \frac{1}{m}(y - y_1)$

$= -8 + \frac{6}{14}(4 - (-8))$

$\therefore x = -8 + \frac{31}{3} = -10.33 \approx -10 \therefore y = \frac{-17}{3}$

$(x', y') = (4, \frac{-17}{3})$

For bit 3, $y = y_{wmin} = -5$

and,

$x = x_1 + \frac{1}{m}(y - y_1)$

$= 4 + \frac{6}{-14}(-5 - (\frac{-17}{3}))$

Available on notes.arpanadhikari7.com.np

Or, $x = 3.71$

$\therefore x \approx 4$

Hence, $(x'', y'') = (4, -5)$

The coordinates of the end points of visible line segment after clipping are A' (-1, 6) and B' (4, -5). Ans

Q7) Derive the composite transformation matrix that reflects an object about line 'L' with necessary figures.

Soln

Let the line 'L' be considered as $y = mx + c$.

Step 1: The line is translated through factor $(0, -c)$ along with the object to be reflected, such that the line passes through origin.

Step 2: Now, the line along with the object is rotated until it aligns with one of the major coordinate axes (consider X-axis) by angle θ.

where, $\theta = \tan^{-1}(m)$ ⟶ value Derived from eq?.

Step 3: The reflection of the object is taken along x-axis.

Step 4: After reflection, the line along with the object is rotated back to the same slope with angle 'θ' but in opposite direction.

Step 5: Now, we perform inverse translation to bring the line back to its initial location along with the reflected object.

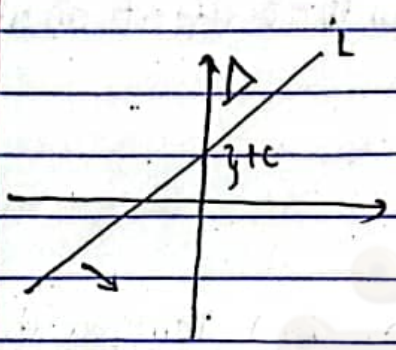
notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

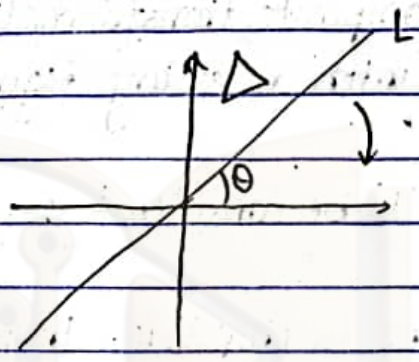
The composite matrix is,

$$C_m = T_{(0,+c)} \cdot R_{\theta cw} \cdot R_{fx} \cdot R_{\theta cw} \cdot T_{(0,-c)}$$

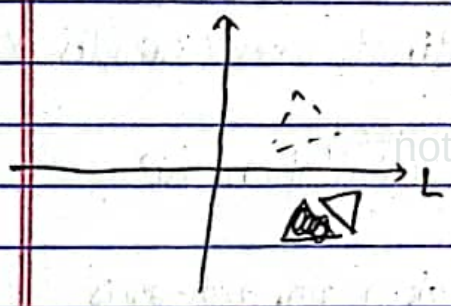
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}$$



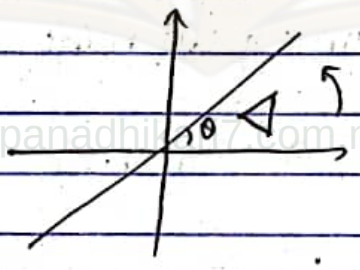
Step : 1 (Translation)



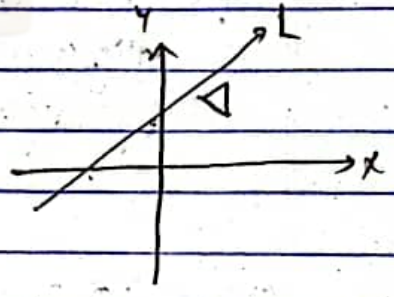
Step 2 : Rotation



step 3 : Reflection



Step 4 : Inverse Rotation



step 5 : Inverse translation

notes.arpanadhikari7.com.np

Q8) Find scaling transformation matrix to scale s_x, s_y units with respect to fixed point P(x,y).

Sol"

Available on notes.arpanadhikari7.com.np

Step 1 : The fixed point P(x,y) along with the object to be rotated about it, is translated to origin.

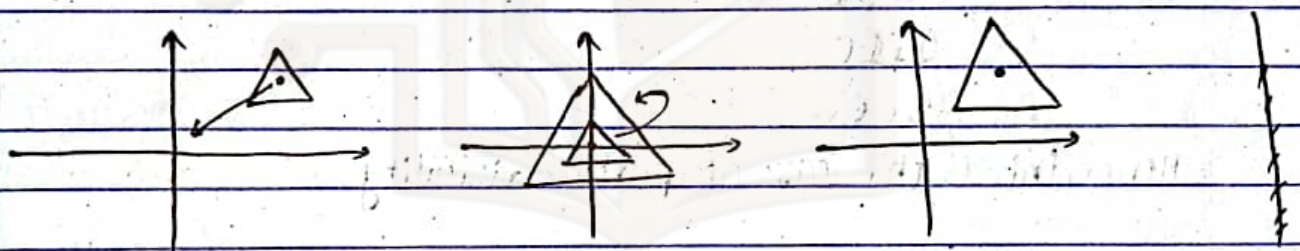
Step 2 : Now scale the object about origin by the desired scaling factor.

Step 3 : Translate back the fixed point which is at the coordinate origin along with the scaled object to its original location.

The composite matrix for these transformations is as follows:

$$CM = T_{(x,y)} \cdot S_{s_x,s_y} \cdot T_{(-x,-y)}$$

$$= \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -x \\ 0 & 1 & -y \\ 0 & 0 & 1 \end{pmatrix}$$



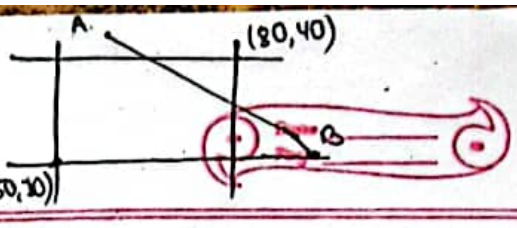
Step: 1 (Translate) : Step2 : scaling Step 3 : Reverse Translation

notes.arpanadhikari7.com.np

Q9) Use cohen sutherland's algorithm to clip line (60,50) (100,10) against window (50,10) (80,40).

soln

Available on notes.arpanadhikari7.com.np



Given points, A(60,50) B(100,10)

$(x_{wmin}, y_{wmin}) = (50, 10)$ $(x_{wmax}, y_{wmax}) = (80, 40)$

Step 1 : Establish region codes for line end points.

A(60,50) :

L : 60 < 50 F 0
R : 60 > 80 F 0
B : 50 < 10 F 0
T : 50 > 40 T 1

Region code ⇒ ~~000~~1000

B(100,10)

L : 100 < 50 F 0
R : 100 > 80 T 1
B : 10 < 10 F 0
T : 10 > 40 F 0

Region code ⇒ ~~0100~~ 0010

Hence, this isn't case of full visibility.

Step 2 : Perform logical ~~AND~~ operation of region codes.

```
  0001
  0010
  ----
  0000
```

Hence, this is the case of partial visibility.

notes.arpanadhikari7.com.np

Step 3 : Determine intersection point of line with window boundaries.

For A, $(x_1, y_1) = (60, 50)$

$y = y_{wmax} = 40$

$$x = \frac{1}{m}(y - y_1) + x_1$$

$$= \frac{1}{-1}(40 - 50) + 60$$

$\therefore x = 70$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10 - 50}{100 - 60} = \frac{-40}{40}$$

$$= -1$$

Available on notes.arpanadhikari7.com.np

For B, $(x_1, y_1) = (100, 10)$

$x = x_{wmax} = 80$

$y = m(x - x_1) + y_1$

$= -1(80 - 100) + 10$

$\therefore y = 30$

Hence, the coordinates of the end points of visible line segment after clipping are ~~A'(~~, A'(70,40) and B'(80,30). Ans

Q10) show that 2D reflection through x-axis followed by 2D reflection through line $y = -x$ is equivalent to a pure rotation (90 degrees) about origin.

Soln

To prove, lets deduce the composite matrix C_{m1} and C_{m2} for two cases.

For first case,

$C_{m1} = R_{fx}$ Given line, $y = -x + 0$

so, $m = -1$

$\therefore \theta = -45°$

composite matrix for line reflection = $R\theta_{cw}^{-45°} \cdot R_{fx} \cdot R\theta_{ccw}^{45°}$

Then, Final C_m with reflection through x-axis is,

$C_{m1} = R\theta_{cw}^{-45°} \cdot R_{fx} \cdot R\theta_{ccw}^{45°} \cdot R_{fx}$

$$= \begin{pmatrix} \cos 45° & \sin 45° & 0 \\ -\sin 45° & \cos 45° & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 45° & -\sin 45° & 0 \\ \sin 45° & \cos 45° & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Available on notes.arpanadhikari7.com.np

So, $C_{m1} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Now, For the second case,

$C_{m2} = R\,\theta_{cw}^{-90^\circ}$

$= \begin{pmatrix} \cos 90^\circ & \sin 90^\circ & 0 \\ -\sin 90^\circ & \cos 90^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

Hence, we can observe $C_{m1} = C_{m2}$

So, 2D reflection through x-axis followed by 2D reflection through line $y = -x$ is equivalent to a pure rotation (90°) about origin.

notes.arpanadhikari7.com.np

Q11) Prove that scaling followed by rotation is equivalent to shearing.

Soln

Available on notes.arpanadhikari7.com.np

Q12) Triangle with vertices A(1,1), B(7,1), C(4,3) is required to be rotated about any arbitrary fixed point (4,2) in a counter clock wise direction by 90 degrees. What will be the final coordinates of the triangle?

Sol'n

Composite Matrix (C_m) = $T_{(4,2)}\ R_{\theta}^{90°}{}_{ccw}\ T_{(-4,-2)}$

$$= \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 90° & -\sin 90° & 0 \\ \sin 90° & \cos 90° & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 & 6 \\ 1 & 0 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

So, Final coordinates, $P' = C_m \cdot P$

$$= \begin{pmatrix} 0 & -1 & 6 \\ 1 & 0 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 7 & 4 \\ 1 & 1 & 3 \\ 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & 5 & 3 \\ -1 & 5 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

Hence, final coordinates triangle after rotation are
A(5,-1), B(5,5), C(3,2). Ans

Q13) A point P(20,20) in a window with its lower most left corner at (10,10) and uppermost right corner at (100,100) is required to be mapped to a point in viewport having its lower most left corner at (30,30) and uppermost right corner at (90,80). Where will

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

the point, P be placed after the transformation?

2017

Given, P(20,20)

$(x_{wmin}, y_{wmin}) = (10,10)$ $(x_{vmin}, y_{vmin}) = (30,30)$

$(x_{wmax}, y_{wmax}) = (100,100)$ $(x_{vmax}, y_{vmax}) = (90,80)$

For this,

Composite matrix = $T_{(x_{vmin}, y_{vmin})} \cdot S_{s_x, s_y} \cdot T_{(x_{wmin}, y_{wmin})}$ ——(i)

Here, $S_x = \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}}$

$= \frac{90-30}{100-10} = \frac{60}{90} = 0.67$

$S_y = \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}}$

$= \frac{80-30}{100-10} = \frac{50}{90} = 0.56$

So, $C_m = \begin{pmatrix} 1 & 0 & 30 \\ 0 & 1 & 30 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0.67 & 0 & 0 \\ 0 & 0.56 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -10 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 0.67 & 0 & 23.3 \\ 0 & 0.56 & 24.4 \\ 0 & 0 & 1 \end{pmatrix}$

So, $P' = C_m \cdot P$

$= \begin{pmatrix} 0.67 & 0 & 23.3 \\ 0 & 0.56 & 24.4 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 20 \\ 20 \\ 1 \end{pmatrix} = \begin{pmatrix} 36.7 \\ 35.6 \\ 1 \end{pmatrix} \approx \begin{pmatrix} 37 \\ 36 \\ 1 \end{pmatrix}$

Hence, the point P will be placed at (37,36) Ans

Available on notes.arpanadhikari7.com.np

Q14) Derive a composite transformation matrix for mapping a point P in a circular window with a radius of 50 pixels and having its center at (50,50) to a circular view port with a radius of 20 pixels and having its center at (10,10).

Sol^n

Here, $(x_{wc}, y_{wc}) = (50,50)$

$(x_{vc}, y_{vc}) = (10,10)$

Then, $W_r = 50$ pixels $\quad V_r = 20$ pixels

So, $S_x = S_y = \frac{V_r}{W_r} = \frac{20}{50} = 0.4$

$$C_m = T(x_{vc}, y_{vc}) \cdot S_{s_x, s_y} \cdot T(-x_{wc}, -y_{wc})$$

$$= \begin{pmatrix} 1 & 0 & 10 \\ 0 & 1 & 10 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0.4 & 0 & 0 \\ 0 & 0.4 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -50 \\ 0 & 1 & -50 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.4 & 0 & -10 \\ 0 & 0.4 & -10 \\ 0 & 0 & 1 \end{pmatrix}_{Ans}$$

notes.arpanadhikari7.com.np

Q15) A window has its lower most left corner at (-5,10) and its upper most right corner at (35,40). Clip a line segment that has two end points (-20,5) and (60,20). If a viewport is defined by (6,17) and (90,80). Find the view port coordinates for the intersection points.

Sol^n

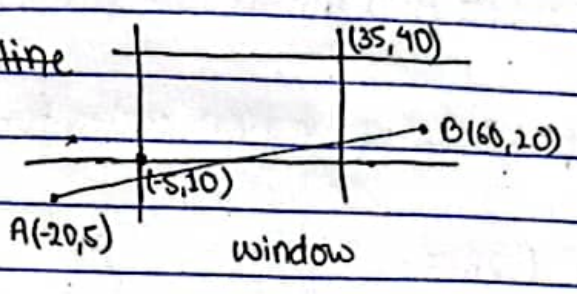
Available on notes.arpanadhikari7.com.np

$(x_{wmin}, y_{wmin}) = (-5, 10)$ $(x_{vmin}, y_{vmin}) = (6, 17)$

$(x_{wmax}, y_{wmax}) = (35, 40)$ $(x_{vmax}, y_{vmax}) = (90, 80)$

End points of line : A (-20, 5) and B (60, 20)

~~Step 1 : Establish region codes for line~~



First, lets ~~translate~~ transform points to viewport.

$$S_x = \frac{x_{vmax} - x_{vmin}}{x_{wmax} - x_{wmin}} = \frac{90 - 6}{35 - (-5)} = 2.1$$

$$S_y = \frac{y_{vmax} - y_{vmin}}{y_{wmax} - y_{wmin}} = \frac{80 - 17}{40 - 10} = 2.1$$

$$C_m = T_{(x_{vmin}, y_{vmin})} \cdot S_{s_x, s_y} \cdot T_{(-x_{wmin}, -y_{wmin})}$$

$$= \begin{pmatrix} 1 & 0 & 6 \\ 0 & 1 & 17 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2.1 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 2.1 & 0 & 16.5 \\ 0 & 2.1 & -4 \\ 0 & 0 & 1 \end{pmatrix}$$

The points mapped to viewport will be A' and B'.

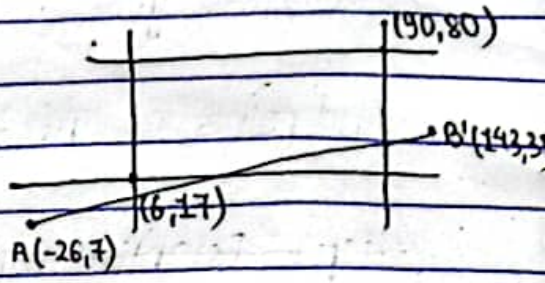
$$P' = C_m \cdot P$$

$$= \begin{pmatrix} 2.1 & 0 & 16.5 \\ 0 & 2.1 & -4 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -20 & 60 \\ 5 & 20 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} -25.5 & 142.5 \\ 6.5 & 38 \\ 1 & 1 \end{pmatrix}$$

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

So, A'(-26.5, 6.5) and B'(142.5, 38)
≈ A'(-26, 7) and B'(143, 38)



Now, Lets find intersection points.

Step1: Develop region codes for line end points

A'(-26,7)			B'(143,38)		
L : -26 < 6	T	1	L : 143 < 6	F	0
R : -26 > 90	F	0	R : 143 > 90	T	1
B : 7 < 17	T	1	B : 38 < 17	F	0
T : 7 > 80	F	0	T : 38 > 80	F	0
Regi		(0101)			(0010)

So, this is not the case of full visibility.

Step 2: Perform AND operation of region code of end points.

$$\begin{array}{r} 1010 \\ 0100 \\ \hline 0000 \end{array}$$

notes.arpanadhikari7.com.np

So, this is the case of partial visibility.

Step 3: Find intersection points for end points.

For B' the slope $(m) = \frac{y_2 - y_1}{x_2 - x_1} = \frac{38-7}{143+26} = \frac{31}{169}$

For B'(143, 38),

$x = x_{vmax} = 90$

$y = m(x - x_1) + y_1$

$= \frac{31}{169}(90 - 143) + 38 \quad \therefore y = 28.27 \approx 28 \quad (90, 28)$

Available on notes.arpanadhikari7.com.np

For A'(-26,7),

For bit 1,

$x = x_{wmin} = 6$

$y = m(x - x_1) + y_1$

$= \frac{31}{169}(6+26) + 7$

$\therefore y = 12.86$ So, $(x', y') = (6, 12.86)$

For bit 3,

$y = y_{wmin} = 17$

$x = \frac{1}{m}(y - y_1') + x_1'$

$= \frac{169}{31}(17 - 12.86) + 6$

$\therefore x = 28.57 \approx 29$ So, (29, 17)

Hence, final coordinates of visible line end points are A(29,17) and B(90,28) Ans

Q16) Reduce a triangle with coordinates A(0,0), B(0,4), C(8,4), D(8,0) to twice its original about a fixed point P(4,2) also derive a transformation matrix to convert this rectangle into a square.

Soln

$$C_m = T_{(4,2)} \cdot S_{S_x=\frac{1}{2}, S_y=\frac{1}{2}} \cdot T(-4,-2)$$

$$= \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

Available on notes.arpanadhikari7.com.np

$$\therefore C_m = \begin{pmatrix} 0.5 & 0 & -2 \\ 0 & 0.5 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$P' = C_m \cdot P$$

$$= \begin{pmatrix} 0.5 & 0 & -2 \\ 0 & 0.5 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 8 & 8 \\ 0 & 4 & 4 & 0 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -4 & 4 & 12 & 12 \\ -2 & 6 & 6 & -2 \\ 1 & 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 6 & 6 \\ 1 & 3 & 3 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$

So, So, rectangle coordinates after reduction are
(2,1), (2,3), (6,3), (6,1). Ans

To convert rectangle to square, Lets calculate 'l' & 'b' for rectangle,

$l = 8 - 0 = 8$

$b = 4 - 0 = 4$

notes.arpanadhikari7.com.np

So, Lets shorten the length from 8 to 4 in x-axis to be square.

So, $S_x = \frac{4}{8} = 0.5$

$S_y = 1$

So, using fined point, we first translate the fined point to centre (0,0), scale by $S_x = 0.5$, $S_y = 1$ and translate back.

The transformation matrix will be,

$$C_m = T_{(+x_f, +y_f)} \cdot S_{s_x = 0.5, s_y = 1} \cdot T_{(-x_f, -y_f)}$$

Available on notes.arpanadhikari7.com.np

$$= \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0.5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.5 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$ Ans

Q17) A triangle with vertices A(0,10), B(10,0), C(-10,0) is required to be shifted down by 5 units, then rotate in anticlockwise direction by 30° and scale by twice its original size. What will be the final location?

Soln

Step 1 : shift down by 5 units i.e. $T_{(-5,-5)}$
Step 2 : Rotate ccw by 30° i.e. $R_{\theta}^{30°}{}_{ccw}$
step 3 : scale by $S_x = 2, S_y = 2$.

So, $C_m = S_{S_x=2, S_y=2} \cdot R_{\theta}^{30°}{}_{ccw} \; T_{(-5,-5)}$

notes.arpanadhikari7.com.np

$$= \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 30° & -\sin 30° & 0 \\ \sin 30° & \cos 30° & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & -5 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1.73 & -1 & -3.66 \\ 1 & 1.73 & -13.66 \\ 0 & 0 & 1 \end{pmatrix}$$

So, $P' = C_m \cdot P$

$$= \begin{pmatrix} 1.73 & -1 & -3.66 \\ 1 & 1.73 & -13.66 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 10 & -10 \\ 10 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

Available on notes.arpanadhikari7.com.np

$$= \begin{pmatrix} -13.66 & 13.66 & -20.98 \\ 3.66 & -3.66 & -23.66 \\ 1 & 1 & 1 \end{pmatrix}$$

$$\approx \begin{pmatrix} 14 & 14 & -21 \\ 4 & -4 & -24 \\ 1 & 1 & 1 \end{pmatrix}$$

Hence, final location will be A(14,4), B(14,-4), C(-21,-24). Ans

Q18) Rotate triangle A(0,0), B(1,1), C(5,2) about origin and about point P(-1,-1) by 45° in a cc direction.

Soln

Composite matrix (C_m) = $T_{(-1,-1)} \cdot R\theta^{45°}_{ccw} \cdot T_{(1,1)} \cdot R\theta^{45°}_{ccw}$

$$= \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos45° & -\sin45° & 0 \\ \sin45° & \cos45° & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos45° & -\sin45° & 0 \\ \sin45° & \cos45° & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & 0.414 \\ 0 & 0 & 1 \end{pmatrix}$$

notes.arpanadhikari7.com.np

$P' = C_m \cdot P$

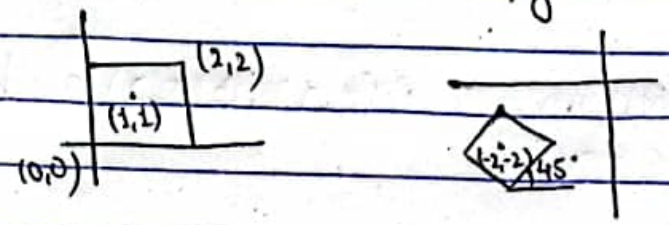
$$= \begin{pmatrix} 0 & -1 & -1 \\ 1 & 0 & 0.414 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 5 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & -2 & -3 \\ 0.414 & 1.414 & 5.414 \\ 1 & 1 & 1 \end{pmatrix}$$

Final points: A'(-1,0), B'(-2,1), C'(-3,5) Ans

Available on notes.arpanadhikari7.com.np

Q19) Write the series of transformation in matrix that are needed to place the square shown in the figure A reduced to one third of its original size, into the position shown in fig. B where the center of square is at (-2,-2).



Sol?

$$C_m = T_{(-2,-2)} \cdot R_{ccw}^{45^\circ} \cdot S_{Sx=\frac{1}{3},Sy=\frac{1}{3}} \cdot T_{(-1,-1)}$$

$$= \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ & 0 \\ \sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1/3 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.236 & -0.236 & -2 \\ 0.236 & 0.236 & -2.47 \\ 0 & 0 & 1 \end{pmatrix}$$

To prove, $P' = C_m \cdot P$

$$= \begin{pmatrix} 0.236 & -0.236 & -2 \\ 0.236 & 0.236 & -2.47 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix} //$$

Hence, series is as:

step 1: Translation by factor (-1, -1) to bring object to origin. (fixed point)

Step 2: scaling by factor 1/3 in both axis.

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Step3: Rotation by 45° in counter clockwise direction.
Step4: Translate ~~to~~ by factor (-2,-2) to move fixed point to req^d point as in fig.

Q20) Reflect a ~~point~~ Triangle with vertices A(4,1), B(5,2) and C(4,3) about line $3x-4y+8=0$.

Sol^n

Given line, $-4y = -3x-8$

or, $y = \frac{3}{4}x + 2$

Here, $m = \frac{3}{4}$; $c = 2$

so, $\theta = \tan^{-1}\left(\frac{3}{4}\right) = 36.87°$

For reflection,

$$C_m = T_{(0,+2)} \cdot R\theta_{(ccw)}^{36.87°} \cdot R_{fx} \cdot R\theta_{cw}^{-36.87°} \cdot T_{(0,-2)}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 36.87 & -\sin 36.87 & 0 \\ \sin 36.87 & \cos 36.87 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 36.87 & \sin 36.87 & 0 \\ -\sin 36.87 & \cos 36.87 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0.279 & 0.960 & -1.92 \\ 0.960 & -0.279 & 0.559 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 0.279 & 0.960 & -1.92 \\ 0.960 & -0.279 & 2.559 \\ 0 & 0 & 1 \end{pmatrix}$$

Available on notes.arpanadhikari7.com.np

So, $P' = C_m \cdot P$

$$= \begin{pmatrix} 0.279 & 0.960 & -1.92 \\ 0.960 & -0.279 & 2.559 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 5 & 4 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{pmatrix}$$

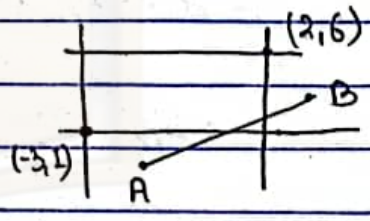
$$= \begin{pmatrix} 0.159 & 1.39 & 2.079 \\ 6.12 & 6.80 & 5.560 \\ 1 & 1 & 1 \end{pmatrix}$$

Hence, Final triangle vertices are approximately,
A(0,6), B(1,7), C(2,6) Ans

Q21) Clip a line with end points A(1,-2) and B(3,3) against a clip window with its lower left corner at (-3,1) and upper right corner at (2,6).

Soln

Line end points : A(1,-2), B(3,3)



Step 1 : Develop region codes for line end points.

notes.arpanadhikari7.com.np

A(1,-2) :

L :	1<-3	F	0
R :	1>2	F	0
B :	-2<1	T	1
T :	-2>6	F	0

Code : (0100)

B(3,3)

L :	3<-3	F	0
R :	3>2	T	1
B :	3<1	F	0
T :	3>6	F	0

Code : (0010)

So, this is not the case of full visibility.

Available on notes.arpanadhikari7.com.np

Step 2 : Perform AND operation of region codes of end points.

$$\begin{array}{r} 0100 \\ 0010 \\ \hline 0000 \end{array}$$

So, this is the case of partial visibility.

Step 3 : Find intersection points with clip window.

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-(-2)}{3-1} = \frac{5}{2}$$

For A, (1, -2)

$y = y_{wmin} = 1$

$x = \frac{1}{m}(y - y_1) + x_1$

$= \frac{2}{5}(1+2) + 1$

$= 2.2 \qquad (2.2, 1) \approx (2, 1)$

For B, (3, 3)

$x = x_{wmax} = 2$

$y = m(x - x_1) + y_1$

$= \frac{5}{2}(2-3) + 3$

$= 0.5 \qquad (2, 0.5) \approx (2, 1)$

Hence, the visible end points of line are A(2, 1) and B(2, 1)
i.e. line passes through intersection point of clip window.

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

Q22) Prove that reflection about line $y=-x$ is equivalent to reflection about y-axis followed by a counter clockwise rotation by 90°.

Ans

Case I : Reflection about line $y=-x$.

Here, $m=-1$

$\therefore \theta = \tan^{-1}(-1)$

$\therefore \theta = -45°$

$$C_m = R\theta^{-45°}_{ccw} \cdot R_{fx} \cdot R\theta^{+45°}_{ccw}$$

$$= \begin{pmatrix} \cos 45° & \sin 45° & 0 \\ -\sin 45° & \cos 45° & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos 45° & -\sin 45° & 0 \\ \sin 45° & \cos 45° & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\therefore C_{m_1} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Case II : counter clockwise rotation by 90° after R_{fy}.

$$C_m = R\theta^{90°}_{ccw} \cdot R_{fy}$$

$$= \begin{pmatrix} \cos 90° & -\sin 90° & 0 \\ \sin 90° & \cos 90° & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\therefore C_{m_2} = \begin{pmatrix} 0 & -1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np

As the composite matrix obtained in both cases are same, reflection about $y = -x$ is equivalent to reflection about y-axis followed by a counter clockwise rotation by 90 degrees.

notes.arpanadhikari7.com.np

Available on notes.arpanadhikari7.com.np